

IPv4 Addressing

1 Introduction to IPv4 Addressing

What is IPv4?

IPv4 (Internet Protocol version 4) is the dominant addressing system used in computer networks. Every device on the Internet must have a unique IPv4 address (unless hidden behind NAT).

Key concepts:

- An IP address identifies a **device interface**, not the device itself.
- Routers have multiple interfaces → each requires its own IP.
- Hosts typically have one interface → one IP.

IPv4 provides:

- Logical addressing (not tied to hardware)
- Routing across networks
- Packet identification and fragmentation

2 IPv4 Address Format

32-bit Structure

IPv4 addresses are **32 bits** long, represented as 4 decimal bytes:

$$a.b.c.d = 32 \text{ bits}$$

Example:

193.32.216.9

Binary representation:

11000001 00100000 11011000 00001001

Total number of IPv4 addresses:

$$2^{32} = 4,294,967,296$$

3 Classful Addressing (Historical System)

Before CIDR, the Internet used **Classful Addressing**, dividing IPv4 into fixed-size blocks.

3.1 Address Classes

Class	Prefix Bits	Address Range	Hosts
A	0xxxxxxx	0.0.0.0 — 127.255.255.255	16,777,214
B	10xxxxxx	128.0.0.0 — 191.255.255.255	65,534
C	110xxxxx	192.0.0.0 — 223.255.255.255	254
D	1110xxxx	224.0.0.0 — 239.255.255.255	Multicast
E	1111xxxx	240.0.0.0 — 255.255.255.255	Reserved

Table 1: Classful IPv4 Addressing System

3.2 Drawbacks of Classful Addressing

- Huge waste of addresses
- Networks were too large or too small
- Routing tables grew unnecessarily

Examples:

- Class A → 16 million hosts (too big for most organizations)
- Class C → 254 hosts (too small for universities/companies)

Because of inefficient usage, Classful addressing was replaced by CIDR in 1993.

4 CIDR — Classless Interdomain Routing

CIDR Notation

CIDR uses:

a.b.c.d/x

where **x** is the prefix length.

4.1 Prefix vs Host bits

<u>Prefix</u>		<u>Host ID</u>
x bits		$32-x$ bits

CIDR allows more flexibility:

- No fixed blocks
- Efficient address utilization
- Reduced global routing table size

5 Subnets and Interfaces

What is a Subnet?

A subnet is a group of interfaces that share the same prefix and are not separated by a router.

Example:

$$223.1.1.0/24 \implies 223.1.1.xxx$$

Subnet Diagrams

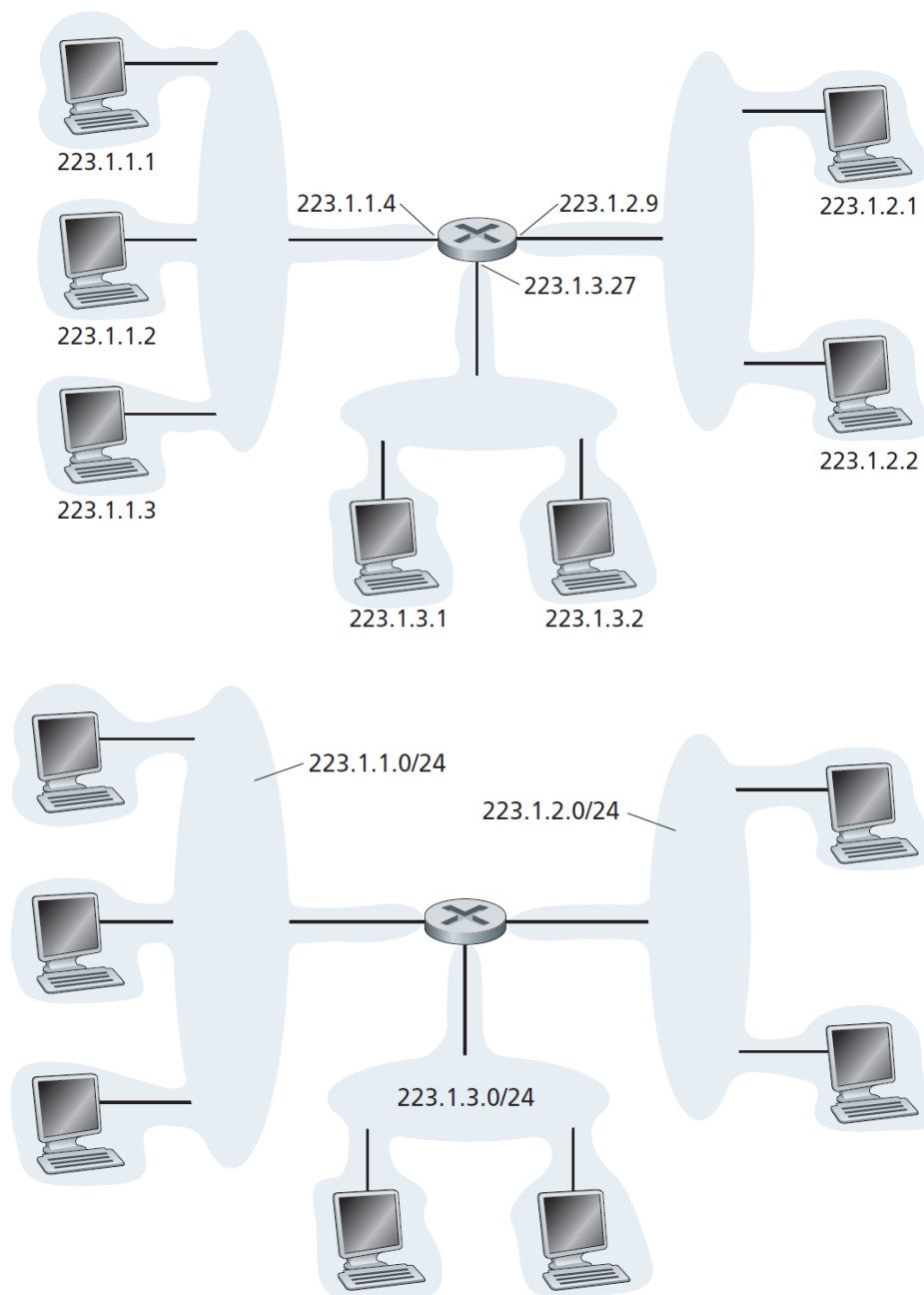


Figure 1: Subnet examples showing hosts and router interfaces

6 Determining Subnets in a Network

General steps:

1. Remove all routers and hosts.
2. Each isolated link or LAN becomes a subnet.

Subnets include:

- LAN segments
- Router-to-router point-to-point links

Example: Multiple Routers Form Multiple Subnets

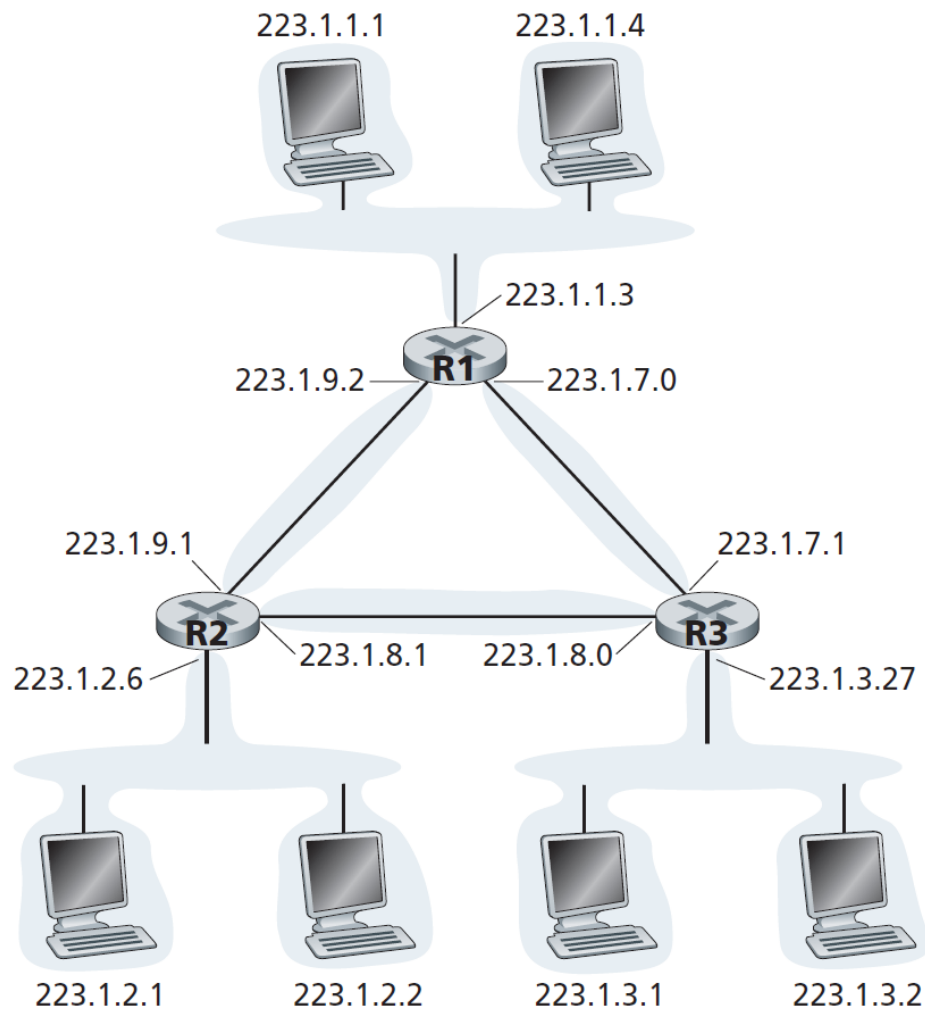


Figure 2: Complex network showing multiple subnets

7 Hierarchical Address Allocation

ISPs receive large IPv4 blocks from ICANN (e.g., /16 or /12). They subdivide these into smaller blocks for organizations.

Example:

200.23.16.0/20

Can be broken into:

8 blocks of size /23

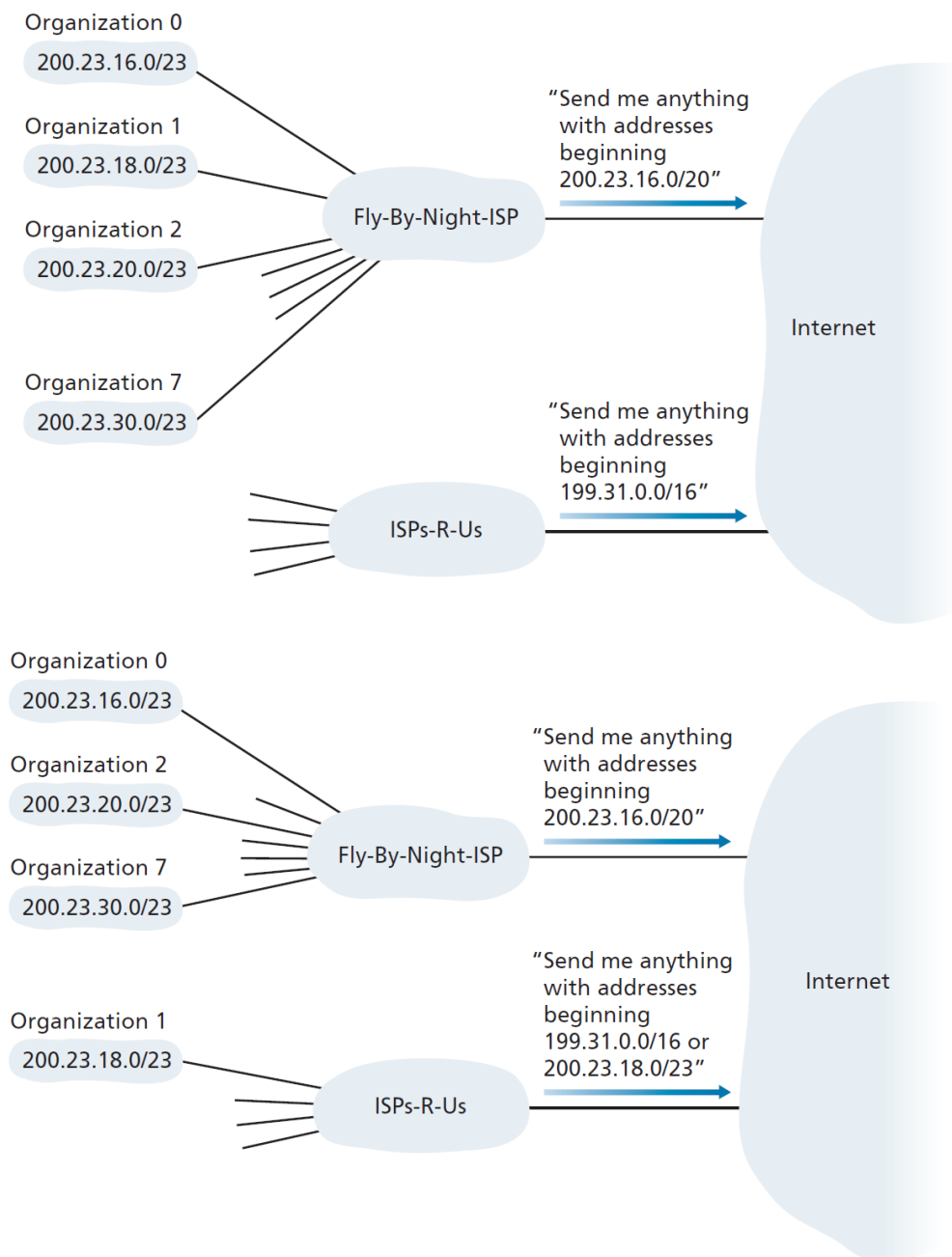


Figure 3: Hierarchical aggregation reduces routing table size

8 Broadcast Addresses

- Limited broadcast: 255.255.255.255
- Directed broadcast example: subnet 192.168.1.0/24

192.168.1.255

Routers do **not** forward broadcast packets.

9 How Organizations Obtain IPv4 Addresses

Two ways:

- From their ISP (most common)
- Directly from ICANN Regional Internet Registries (RIRs):
 - ARIN (North America)
 - RIPE (Europe)
 - APNIC (Asia Pacific)
 - LACNIC (Latin America)
 - AfriNIC (Africa)

ICANN manages:

- Global IP address allocation
- DNS root zone
- Top-level domain distribution